

19

Series and Residues

EXERCISES 19.1

Sequences and Series

1. $5i, -5, -5i, 5, 5i$
2. $2 - i, 1, 2 + i, 3, 2 - i$
3. $0, 2, 0, 2, 0$
4. $1 + i, 2i, -2 + 2i, -4, -4 - 4i$
5. Converges. To see this write the general term as $\frac{3i + 2/n}{1 + i}$.
6. Converges. To see this write the general term as $\left(\frac{2}{5}\right)^n \frac{1 + n2^{-n}i}{1 + 3n5^{-n}i}$.
7. Converges. To see this write the general term as $\frac{(i + 2/n)^2}{i}$.
8. Diverges. To see this consider the term $\frac{n}{n+1}i^n$ and take n to be an odd positive integer.
9. Diverges. To see this write the general term as $\sqrt{n} \left(1 + \frac{1}{\sqrt{n}}i^n\right)$.
10. Converges. The real part of the general term converges to 0 and the imaginary part of the general term converges to π .
11. $\operatorname{Re}(z_n) = \frac{8n^2 + n}{4n^2 + 1} \rightarrow 2$ as $n \rightarrow \infty$, and $\operatorname{Im}(z_n) = \frac{6n^2 - 4n}{4n^2 + 1} \rightarrow \frac{3}{2}$ as $n \rightarrow \infty$.
12. Write $z_n = \left(\frac{1}{4} + \frac{1}{4}i\right)^n$ in polar form as $z_n = \left(\frac{\sqrt{2}}{4}\right)^n \cos n\theta + i \left(\frac{\sqrt{2}}{4}\right)^n \sin n\theta$. Now

$$\operatorname{Re}(z_n) = \left(\frac{\sqrt{2}}{4}\right)^n \cos n\theta \rightarrow 0 \text{ as } n \rightarrow \infty \quad \text{and} \quad \operatorname{Im}(z_n) = \left(\frac{\sqrt{2}}{4}\right)^n \sin n\theta \rightarrow 0 \text{ as } n \rightarrow \infty$$

since $\sqrt{2}/4 < 1$.

13. $S_n = \frac{1}{1+2i} - \frac{1}{2+2i} + \frac{1}{2+2i} - \frac{1}{3+2i} + \frac{1}{3+2i} - \frac{1}{4+2i} + \cdots + \frac{1}{n+2i} - \frac{1}{n+1+2i} = \frac{1}{1+2i} - \frac{1}{n+1+2i}$
Thus, $\lim_{n \rightarrow \infty} S_n = \frac{1}{1+2i} = \frac{1}{5} - \frac{2}{5}i$.

14. By partial fractions, $\frac{i}{k(k+1)} = \frac{i}{k} - \frac{i}{k+1}$ and so

$$S_n = i - \frac{i}{2} + \frac{i}{2} - \frac{i}{3} + \frac{i}{3} - \frac{i}{4} + \cdots + \frac{i}{n} - \frac{i}{n+1} = i - \frac{i}{n+1}.$$

Thus $\lim_{n \rightarrow \infty} S_n = i$.

15. We identify $a = 1$ and $z = 1 - i$. Since $|z| = \sqrt{2} > 1$ the series is divergent.

16. We identify $a = 4i$ and $z = 1/3$. Since $|z| = 1/3 < 1$ the series converges to

$$\frac{4i}{1 - 1/3} = 6i.$$

17. We identify $a = i/2$ and $z = i/2$. Since $|z| = 1/2 < 1$ the series converges to

$$\frac{i/2}{1 - i/2} = -\frac{1}{5} + \frac{2}{5}i.$$

18. We identify $a = 1/2$ and $z = i$. Since $|z| > 1$ the series is divergent.

19. We identify $a = 3$ and $z = 2/(1 + 2i)$. Since $|z| = 2/\sqrt{5} < 1$ the series converges to

$$\frac{3}{1 - \frac{2}{1 + 2i}} = \frac{9}{5} - \frac{12}{5}i.$$

20. We identify $a = -1/(1 + i)$ and $z = i/(1 + i)$. Since $|z| = 1/\sqrt{2} < 1$ the series converges to

$$\frac{-\frac{1}{1+i}}{1 - \frac{i}{1+i}} = -1.$$

21. From
$$\lim_{n \rightarrow \infty} \left| \frac{\frac{1}{(1 - 2i)^{n+2}}}{\frac{1}{(1 - 2i)^{n+1}}} \right| = \frac{1}{|1 - 2i|} = \frac{1}{\sqrt{5}}$$

we see that the radius of convergence is $R = \sqrt{5}$. The circle of convergence is $|z - 2i| = \sqrt{5}$.

22. From
$$\lim_{n \rightarrow \infty} \left| \frac{\frac{1}{n+1} \left(\frac{i}{1+i} \right)^{n+1}}{\frac{1}{n} \left(\frac{i}{1+i} \right)^n} \right| = \lim_{n \rightarrow \infty} \frac{n}{n+1} \left| \frac{i}{1+i} \right| = \frac{1}{\sqrt{2}}$$

we see that the radius of convergence is $R = \sqrt{2}$. The circle of convergence is $|z| = \sqrt{2}$.

23. From
$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(-1)^{n+1}}{(n+1)2^{n+1}}}{\frac{(-1)^n}{n2^n}} \right| = \lim_{n \rightarrow \infty} \frac{n}{2(n+1)} = \frac{1}{2}$$

we see that the radius of convergence is $R = 2$. The circle of convergence is $|z - 1 - i| = 2$.

24. From
$$\lim_{n \rightarrow \infty} \left| \frac{\frac{1}{(n+1)^2(3+4i)^{n+1}}}{\frac{1}{n^2(3+4i)^n}} \right| = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^2 \frac{1}{|3+4i|} = \frac{1}{5}$$

we see that the radius of convergence is $R = 5$. The circle of convergence is $|z + 3i| = 5$.

25. From
$$\lim_{n \rightarrow \infty} \sqrt[n]{|1 + 3i|^n} = |1 + 3i| = \sqrt{10}$$

we see that the radius of convergence is $R = 1/\sqrt{10}$. The circle of convergence is $|z - i| = 1/\sqrt{10}$.

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26. From
$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{1}{n^n} \right|} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

we see that the radius of convergence is ∞ . The power series with center 0 converges absolutely for all z .

27. From
$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{1}{5^{2n}} \right|} = \lim_{n \rightarrow \infty} \frac{1}{25} = \frac{1}{25}$$

we see that the radius of convergence is $R = 25$. The circle of convergence is $|z - 4 - 3i| = 25$.

28. From
$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| (-1)^n \left(\frac{1+2i}{2} \right)^n \right|} = \lim_{n \rightarrow \infty} \left| \frac{1+2i}{2} \right| = \frac{\sqrt{5}}{2}$$

we see that the radius of convergence is $R = 2/\sqrt{5}$. The circle of convergence is $|z + 2i| = 2/\sqrt{5}$.

29. The circle of convergence is $|z - i| = 2$. Since the series of absolute values

$$\sum_{k=1}^{\infty} \left| \frac{(z-1)^k}{k2^k} \right| = \sum_{k=1}^{\infty} \frac{|z-1|^k}{k2^k} = \sum_{k=1}^{\infty} \frac{2^k}{k2^k} = \sum_{k=1}^{\infty} \frac{1}{k}$$

is the divergent harmonic series. But $z = -2 + i$ is on the circle of convergence and $(z - i)^k = (-2)^k$. The series

$$\sum_{k=1}^{\infty} \frac{(-2)^k}{k2^k} = \sum_{k=1}^{\infty} \frac{(-1)^k}{k}$$

is convergent.

30. (a) The circle of convergence is $|z| = 1$. Since the series of absolute values

$$\sum_{k=1}^{\infty} \left| \frac{z^k}{k^2} \right| = \sum_{k=1}^{\infty} \frac{|z|^k}{k^2} = \sum_{k=1}^{\infty} \frac{1}{k^2}$$

converges, the given series is absolutely convergent for every z on $|z| = 1$. Since absolute convergence implies convergence, the given series converges for all z on $|z| = 1$.

(b) The circle of convergence is $|z| = 1$. On the circle, $n|z|^n \rightarrow \infty$ as $n \rightarrow \infty$. This implies $nz^n \not\rightarrow 0$ as $n \rightarrow \infty$. Thus by Theorem 19.3 the series is divergent for every z on the circle $|z| = 1$.

EXERCISES 19.2

Taylor Series

1. $\frac{z}{1+z} = z[1 - z + z^2 - z^3 + \cdots] = z - z^2 + z^3 - z^4 + \cdots = \sum_{k=1}^{\infty} (-1)^{k+1} z^k; \quad R = 1$

2. $\frac{1}{4-2z} = \frac{1}{4} \cdot \left[1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} + \cdots \right] = \frac{1}{4} \sum_{k=0}^{\infty} \frac{z^k}{2^k}; \quad R = 2$

3. Differentiating $\frac{1}{1+2z} = 1 - 2z + 2^2z^2 - 2^3z^3 + \cdots$ gives $\frac{-2}{(1+2z)^2} = -2 + 2 \cdot 2^2z - 3 \cdot 2^3z^2 + \cdots$. Thus

$$\frac{1}{(1+2z)} = 1 - 2 \cdot (2z) + 3 \cdot (2z)^2 - \cdots = \sum_{k=1}^{\infty} (-1)^{k-1} k (2z)^{k-1} \text{ where } R = \frac{1}{2}.$$

4. Using the binomial series gives

$$\frac{z}{(1-z)} = z \left[1 + 3z + \frac{3 \cdot 4}{2!} z^2 + \frac{3 \cdot 4 \cdot 5}{3!} z^3 + \cdots \right] = z + 3z^2 + \frac{3 \cdot 4}{2!} z^3 + \frac{3 \cdot 4 \cdot 5}{3!} z^4 + \cdots \text{ where } R = 1.$$

5. Replacing z in $e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}$ by $-2z$ gives $e^{-2z} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} (2z)^k$ where $R = \infty$.

6. Replacing z in $e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}$ by $-z^2$ and multiplying the result by z gives $ze^{-z^2} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} z^{2k+1}$ where $R = \infty$.

7. Subtracting the series for e^z and e^{-z} gives $\sinh z = \frac{1}{2}(e^z - e^{-z}) = \sum_{k=0}^{\infty} \frac{z^{2k+1}}{(2k+1)!}$ where $R = \infty$.

8. Adding the series for e^z and e^{-z} gives $\cosh z = \frac{1}{2}(e^z + e^{-z}) = \sum_{k=0}^{\infty} \frac{z^{2k}}{(2k)!}$ where $R = \infty$.

9. Replacing z in $\cos z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!}$ by $z/2$ gives $\cos \frac{z}{2} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} \left(\frac{z}{2}\right)^{2k}$ where $R = \infty$.

10. Replacing z in $\sin z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!}$ by $3z$ gives $\sin 3z = \sum_{k=0}^{\infty} (-1)^k \frac{(3z)^{2k+1}}{(2k+1)!}$ where $R = \infty$.

11. Replacing z in $\sin z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!}$ by z^2 gives $\sin z^2 = \sum_{k=0}^{\infty} (-1)^k \frac{z^{4k+2}}{(2k+1)!}$ where $R = \infty$.

12. Using the identity $\cos z = \frac{1}{2}(1 + \cos 2z)$ and the series $\cos z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!}$ gives

$$\cos^2 z = \frac{1}{2} + \frac{1}{2} \sum_{k=0}^{\infty} (-1)^k \frac{(2z)^{2k}}{(2k)!} = 1 + \sum_{k=1}^{\infty} (-1)^k \frac{2^{2k-1}}{(2k)!} z^{2k} \text{ where } R = \infty.$$

13. Using (6) of Section 19.1,

$$\frac{1}{z} = \frac{1}{1 + (z-1)} = 1 - (z-1) + (z-1)^2 - (z-1)^3 + \cdots = \sum_{k=0}^{\infty} (-1)^k (z-1)^k \text{ where } R = 1.$$

14. Using (6) of Section 19.1,

$$\begin{aligned} \frac{1}{z} &= \frac{1}{1+i+(z-1-i)} = \frac{1}{1+i} \cdot \frac{1}{1+\frac{z-1-i}{1+i}} = \frac{1}{1+i} \left[1 - \frac{(z-1-i)}{1+i} + \frac{(z-1-i)^2}{(1+i)^2} - \frac{(z-1-i)^3}{(1+i)^3} + \cdots \right] \\ &= \frac{1}{1+i} - \frac{(z-1-i)}{(1+i)^2} + \frac{(z-1-i)^2}{(1+i)^3} - \frac{(z-1-i)^3}{(1+i)^4} + \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{(z-1-i)^k}{(1+i)^{k+1}} \text{ where } R = \sqrt{2} \end{aligned}$$

15. Using (5) of Section 19.1,

$$\begin{aligned} \frac{1}{3-z} &= \frac{1}{3-2i-(z-2i)} = \frac{1}{3-2i} \cdot \frac{1}{1-\frac{z-2i}{3-2i}} \\ &= \frac{1}{3-2i} \left[1 + \frac{z-2i}{3-2i} + \frac{(z-2i)^2}{(3-2i)^2} + \frac{(z-2i)^3}{(3-2i)^3} + \cdots \right] \\ &= \frac{1}{3-2i} + \frac{z-2i}{(3-2i)^2} + \frac{(z-2i)^2}{(3-2i)^3} + \frac{(z-2i)^3}{(3-2i)^4} + \cdots = \sum_{k=0}^{\infty} \frac{(z-2i)^k}{(3-2i)^{k+1}} \text{ where } R = \sqrt{13}. \end{aligned}$$

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16. Using (6) of Section 19.1,

$$\begin{aligned}\frac{1}{1+z} &= \frac{1}{1-i+z+i} = \frac{1}{1-i} \cdot \frac{1}{1+\frac{z+i}{1-i}} = \frac{1}{1-i} \left[1 - \frac{z+i}{1-i} + \frac{(z+i)^2}{(1-i)^2} - \frac{(z+i)^3}{(1-i)^3} + \cdots \right] \\ &= \frac{1}{1-i} - \frac{z+i}{(1-i)^2} + \frac{(z+i)^2}{(1-i)^3} - \frac{(z+i)^3}{(1-i)^4} + \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{(z+i)^k}{(1-i)^{k+1}} \quad \text{where } R = \sqrt{2}.\end{aligned}$$

17. Using (5) of Section 19.1,

$$\begin{aligned}\frac{z-1}{3-z} &= (z-1) \cdot \frac{1}{2-(z-1)} = \frac{(z-1)}{2} \cdot \frac{1}{1-\frac{z-1}{2}} = \frac{z-1}{2} \left[1 + \frac{z-1}{2} + \frac{(z-1)^2}{2^2} + \frac{(z-1)^3}{2^3} + \cdots \right] \\ &= \frac{z-1}{2} + \frac{(z-1)^2}{2^2} + \frac{(z-1)^3}{2^3} + \frac{(z-1)^4}{2^4} + \cdots = \sum_{k=1}^{\infty} \frac{(z-1)^k}{2^k} \quad \text{where } R = 2.\end{aligned}$$

18. Using (5) of Section 19.1,

$$\begin{aligned}\frac{1+z}{1-z} &= -1 + \frac{2}{1-z} = -1 + \frac{2}{1-i-(z-i)} = -1 + \frac{2}{1-i} \cdot \frac{1}{1-\frac{z-i}{1-i}} \\ &= -1 + \frac{2}{1-i} \left[1 + \frac{z-i}{1-i} + \frac{(z-i)^2}{(1-i)^2} + \frac{(z-i)^3}{(1-i)^3} + \cdots \right] \\ &= -1 + \frac{2}{1-i} + \frac{2(z-i)}{(1-i)^2} + \frac{2(z-i)^2}{(1-i)^3} + \frac{2(z-i)^3}{(1-i)^4} + \cdots = -1 + \sum_{k=0}^{\infty} \frac{2(z-i)^k}{(1-i)^{k+1}} \quad \text{where } R = \sqrt{2}.\end{aligned}$$

19. Using (8) of Section 19.2,

$$\cos z = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2 \cdot 1!} \left(z - \frac{\pi}{4} \right) - \frac{\sqrt{2}}{2 \cdot 2!} \left(z - \frac{\pi}{4} \right)^2 + \frac{\sqrt{2}}{2 \cdot 3!} \left(z - \frac{\pi}{4} \right)^3 + \cdots \quad \text{where } R = \infty.$$

20. Using the identity $\sin z = \cos(z - \pi/2)$ and (14) of Section 19.2, $\sin z = \sum_{k=0}^{\infty} (-1)^k \frac{(z - \frac{\pi}{2})^{2k}}{(2k)!}$ where $R = \infty$.

21. Using $e^z = e^{3i} \cdot e^{z-3i}$ and (12) of Section 19.2, $e^z = e^{3i} \sum_{k=0}^{\infty} \frac{(z-3i)^k}{k!}$ where $R = \infty$.

22. Using $(z-1)e^{-2z} = e^2(z-1)e^{-2(z-1)}$ and (12) of Section 19.2,

$$(z-1)e^{-2z} = e^2 \sum_{k=0}^{\infty} \frac{(-1)^k 2^k}{k!} (z-1)^{k+1} \quad \text{where } R = \infty.$$

23. Using (8) of Section 19.2, $\tan z = z + \frac{1}{3}z^3 + \frac{2}{15}z^5 + \cdots$.

24. Using (8) of Section 19.2, $e^{1/(1+z)} = e - ez + \frac{3e}{2}z^2 - \cdots$.

25. Using (5) of Section 19.1,

$$\begin{aligned}f(z) &= \frac{1}{z-2i} - \frac{1}{z-i} = -\frac{1}{2i} \cdot \frac{1}{1-z/2i} + \frac{1}{i} \cdot \frac{1}{1-z/i} \\ &= -\frac{1}{2i} \left(1 + \frac{z}{2i} + \frac{z^2}{(2i)^2} + \frac{z^3}{(2i)^3} + \cdots \right) + \frac{1}{i} \left(1 + \frac{z}{i} + \frac{z^2}{i^2} + \frac{z^3}{i^3} + \cdots \right) = -\frac{i}{2} - \frac{3}{4}z + \frac{7i}{8}z^2 + \frac{15}{16}z^3 - \cdots.\end{aligned}$$

The radius of convergence is $R = 1$.

26. Using (6) and (5), respectively, of Section 19.1,

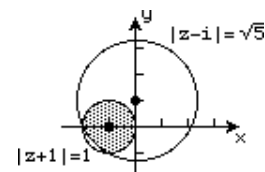
$$\begin{aligned} f(z) &= \frac{2}{z+1} - \frac{1}{z-3} = 2 \cdot \frac{1}{1+z} + \frac{1}{3} \cdot \frac{1}{1-z/3} = 2(1-z+z^2-z^3+\cdots) + \frac{1}{3} \left(1 + \frac{z}{3} + \frac{z^2}{3^2} + \frac{z^3}{3^3} + \cdots \right) \\ &= \frac{7}{3} - \frac{17}{9}z + \frac{55}{27}z^2 - \frac{161}{81}z^3 + \cdots. \end{aligned}$$

27. The distance from $2+5i$ to i is $|2+5i-i| = |2+4i| = 2\sqrt{5}$.

28. The distance from πi to 0 is $|\pi i| = \pi$.

29. The Taylor series are

$$f(z) = \sum_{k=0}^{\infty} (-1)^k (z+1)^k \text{ where } R=1; \text{ and } f(z) = \sum_{k=0}^{\infty} (-1)^k \frac{(z-i)^k}{(2+i)^{k+1}} \text{ where } R=\sqrt{5}.$$

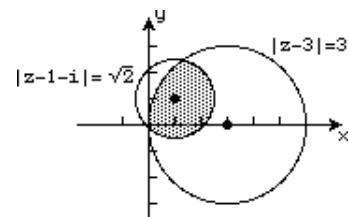


30. The series are

$$f(z) = \sum_{k=0}^{\infty} (-1)^k \frac{(z-3)^k}{3^{k+1}} \text{ where } R=3$$

and

$$f(z) = \sum_{k=0}^{\infty} (-1)^k \frac{(z-1-i)^k}{(1+i)^{k+1}} \text{ where } R=\sqrt{2}.$$



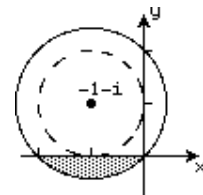
31. (a) The distance from z_0 to the branch cut is one unit.

- (b) The first term of the series determined by Taylor's Theorem is

$$f(-1+i) = \text{Ln}(-1+i) = \log_e \sqrt{2} + \frac{3\pi}{4}i = \frac{1}{2} \log_e 2 + \frac{3\pi}{4}i.$$

The subsequent terms of the series come from $f'(z) = \frac{1}{z}$, $f''(z) = -\frac{1}{z^2}$, and so on, evaluated at $-1+i$.

- (c) The series converges within the circle $|z+1-i| = \sqrt{2}$. Although the series converges in the shaded region, it does not converge to (or represent) $\text{Ln } z$ in this region.



32. (a) $R=1$, which is the distance from the origin to $z=-1$.

- (b) Using Taylor's Theorem [or integrating the series for $1/(1+z)$] we obtain for $R=1$,

$$\text{Ln}(1+z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} z^k.$$

- (c) By replacing z in part (b) by $-z$ we obtain for $R=1$,

$$\text{Ln}(1-z) = -\sum_{k=0}^{\infty} \frac{z^k}{k}.$$

- (d) One way of obtaining the Maclaurin series for $\text{Ln} \left(\frac{1+z}{1-z} \right)$ is to use Taylor's Theorem. Alternatively, let us write

$$\text{Ln} \left(\frac{1+z}{1-z} \right) = \text{Ln}(1+z) - \text{Ln}(1-z)$$

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and subtract the series in parts (b) and (c). This gives for the common circle of convergence $|z| = 1$,

$$\operatorname{Ln} \left(\frac{1+z}{1-z} \right) = 2z + \frac{2}{3}z^3 + \frac{2}{5}z^5 + \frac{2}{7}z^7 + \cdots = 2 \sum_{k=0}^{\infty} \frac{1}{(2k+1)} z^{2k+1}.$$

But recall that in general $\operatorname{Ln}(z_1/z_2) \neq \operatorname{Ln}z_1 - \operatorname{Ln}z_2$ since $\operatorname{Ln}z_1$ and $\operatorname{Ln}z_2$ could differ by a constant multiple of i . That is, $\operatorname{Ln}z_1 - \operatorname{Ln}z_2 = Ci$ for some C . So

$$\operatorname{Ln} \left(\frac{1+z}{1-z} \right) = \operatorname{Ln}(1+z) - \operatorname{Ln}(1-z) - Ci.$$

When $z = 0$ we obtain $\operatorname{Ln} 1 = \operatorname{Ln} 1 - \operatorname{Ln} 1 - Ci$. Since $\operatorname{Ln} 1 = 0$ we get $C = 0$.

33. From $e^z \approx 1 + z + \frac{z^2}{2}$ we obtain

$$e^{(1+i)/10} \approx 1 + \frac{1+i}{10} + \frac{(1+i)^2}{100} = 1.1 + 0.12i.$$

34. From $\sin z \approx z - \frac{z^3}{6}$ we obtain

$$\sin \left(\frac{1+i}{10} \right) \approx \frac{1+i}{10} - \frac{1}{6} \left(\frac{1+i}{10} \right)^3 = \frac{1}{10} + \frac{1}{10}i - \frac{1}{6} \left(\frac{-2+2i}{1000} \right) = \frac{301}{3000} + \frac{299}{3000}i.$$

35. Using the series $e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}$ we obtain $e^{-t^2} = \sum_{k=0}^{\infty} (-1)^k \frac{t^{2k}}{k!}$. Thus

$$\frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt = \frac{2}{\sqrt{\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \int_0^z t^{2k} dt = \frac{2}{\sqrt{\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(2k+1)} z^{2k+1}.$$

$$\begin{aligned} 36. \quad e^{iz} &= \sum_{k=0}^{\infty} \frac{(iz)^k}{k!} = 1 + i \frac{z}{1!} - \frac{z^2}{2!} - i \frac{z^3}{3!} + \frac{z^4}{4!} + i \frac{z^5}{5!} - \frac{z^6}{6!} - i \frac{z^7}{7!} + \cdots \\ &= \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots \right) + i \left(\frac{z}{1!} - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots \right) = \cos z + i \sin z \end{aligned}$$

EXERCISES 19.3

Laurent Series

1. $f(z) = \frac{1}{z} \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots \right) = \frac{1}{z} - \frac{z}{2!} + \frac{z^3}{4!} - \frac{z^5}{6!} + \cdots$
2. $f(z) = \frac{1}{z^5} \left[z - \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots \right) \right] = \frac{1}{3!z^2} - \frac{1}{5!} + \frac{z^2}{7!} - \frac{z^4}{9!} + \cdots$
3. $f(z) = 1 - \frac{1}{1!z^2} + \frac{1}{2!z^4} - \frac{1}{3!z^6} + \cdots$
4. $f(z) = \frac{1}{z^2} \left[1 - \left(1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots \right) \right] = -\frac{1}{1!z} - \frac{1}{2!} - \frac{z}{3!} - \frac{z^2}{4!} - \cdots$

$$5. f(z) = \frac{e \cdot e^{z-1}}{z-1} = \frac{e}{z-1} \left(1 + \frac{(z-1)}{1!} + \frac{(z-1)^2}{2!} + \frac{(z-1)^3}{3!} + \cdots \right) = \frac{e}{z-1} + \frac{e}{1!} + \frac{e(z-1)}{2!} + \frac{e(z-1)^2}{3!} + \cdots$$

$$6. f(z) = z \left(1 - \frac{1}{2!z^2} + \frac{1}{4!z^4} - \frac{1}{6!z^6} + \cdots \right) = z - \frac{1}{2!z} + \frac{1}{4!z^3} - \frac{1}{6!z^5} + \cdots$$

$$7. f(z) = -\frac{1}{3z} \cdot \frac{1}{1 - \frac{z}{3}} = -\frac{1}{3z} \left[1 + \frac{z}{3} + \frac{z^2}{3^2} + \frac{z^3}{3^3} + \cdots \right] = -\frac{1}{3z} - \frac{1}{3^2} - \frac{z}{3^3} - \frac{z^2}{3^4} - \cdots$$

$$8. f(z) = \frac{1}{z^2} \cdot \frac{1}{1 - \frac{3}{z}} = \frac{1}{z^2} \left[1 + \frac{3}{z} + \frac{3^2}{z^2} + \frac{3^3}{z^3} + \cdots \right] = \frac{1}{z^2} + \frac{3}{z^3} + \frac{3^2}{z^4} + \frac{3^3}{z^5} + \cdots$$

$$9. f(z) = \frac{1}{z-3} \cdot \frac{1}{3+z-3} = \frac{1}{3(z-3)} \cdot \frac{1}{1 + \frac{z-3}{3}} = \frac{1}{3(z-3)} \left[1 - \frac{z-3}{3} + \frac{(z-3)^2}{3^2} - \frac{(z-3)^3}{3^3} + \cdots \right]$$

$$= \frac{1}{3(z-3)} - \frac{1}{3^2} + \frac{z-3}{3^3} - \frac{(z-3)^2}{3^4} + \cdots$$

$$10. f(z) = \frac{1}{z-3} \cdot \frac{1}{z-3+3} = \frac{1}{(z-3)^2} \cdot \frac{1}{1 + \frac{3}{z-3}} = \frac{1}{(z-3)^2} \left[1 - \frac{3}{z-3} + \frac{3^2}{(z-3)^2} - \frac{3^3}{(z-3)^3} + \cdots \right]$$

$$= \frac{1}{(z-3)^2} - \frac{3}{(z-3)^3} + \frac{3^2}{(z-3)^4} - \frac{3^3}{(z-3)^5} + \cdots$$

$$11. f(z) = \frac{1}{3} \left[\frac{1}{z-3} - \frac{1}{z} \right] = \frac{1}{3} \left[\frac{1}{z-4+1} - \frac{1}{4+z-4} \right] = \frac{1}{3} \left[\frac{1}{z-4} \cdot \frac{1}{1 + \frac{1}{z-4}} - \frac{1}{4} \cdot \frac{1}{1 + \frac{1}{4}} \right]$$

$$= \frac{1}{3} \left[\frac{1}{z-4} \left(1 - \frac{1}{z-4} + \frac{1}{(z-4)^2} - \frac{1}{(z-4)^3} + \cdots \right) - \frac{1}{4} \left(1 - \frac{z-4}{4} + \frac{(z-4)^2}{4^2} - \frac{(z-4)^3}{4^3} + \cdots \right) \right]$$

$$= \cdots - \frac{1}{3(z-4)^2} + \frac{1}{3(z-4)} - \frac{1}{12} + \frac{z-4}{3 \cdot 4^2} - \frac{(z-4)^2}{3 \cdot 4^3} + \cdots$$

$$12. f(z) = \frac{1}{3} \left[\frac{1}{z-3} - \frac{1}{z} \right] = \frac{1}{3} \left[\frac{1}{-4+z+1} - \frac{1}{z+1-1} \right] = \frac{1}{3} \left[-\frac{1}{4} \cdot \frac{1}{1 - \frac{z+1}{4}} - \frac{1}{z+1} \cdot \frac{1}{1 - \frac{1}{z+1}} \right]$$

$$= \frac{1}{3} \left[-\frac{1}{4} \left(1 + \frac{z+1}{4} + \frac{(z+1)^2}{4^2} + \frac{(z+1)^3}{4^3} + \cdots \right) - \frac{1}{z+1} \left(1 + \frac{1}{z+1} + \frac{1}{(z+1)^2} + \frac{1}{(z+1)^3} + \cdots \right) \right]$$

$$= \cdots - \frac{1}{(z+1)^2} - \frac{1}{z+1} - \frac{1}{12} - \frac{z+1}{3 \cdot 4^2} - \frac{(z+1)^2}{3 \cdot 4^3} - \cdots$$

$$13. f(z) = \frac{1}{z-2} - \frac{1}{z-1} = -\frac{1}{2} \cdot \frac{1}{1 - \frac{z}{2}} - \frac{1}{z} \cdot \frac{1}{1 - \frac{1}{z}}$$

$$= -\frac{1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} + \cdots \right) - \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots \right) = \cdots - \frac{1}{z^2} - \frac{1}{z} - \frac{1}{2} - \frac{z}{2^2} - \frac{z^2}{2^3} - \cdots$$

$$14. f(z) = \frac{1}{z-2} - \frac{1}{z-1} = \frac{1}{z} \cdot \frac{1}{1 - \frac{2}{z}} - \frac{1}{z} \cdot \frac{1}{1 - \frac{1}{z}} = \frac{1}{z} \left(1 + \frac{2}{z} + \frac{2^2}{z^2} + \frac{2^3}{z^3} + \cdots \right) - \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots \right)$$

$$= \frac{1}{z^2} + \frac{2^2-1}{z^3} + \frac{2^3-1}{z^4} + \frac{2^4-1}{z^5} + \cdots$$

19.3 Laurent Series

$$15. f(z) = \frac{1}{z-1} \cdot \frac{-1}{1-(z-1)} = \frac{-1}{z-1} [1 + (z-1) + (z-1)^2 + (z-1)^3 + \cdots] = -\frac{1}{z-1} - 1 - (z-1) - (z-1)^2 - \cdots$$

$$16. f(z) = \frac{1}{z-2} \cdot \frac{1}{1+(z-2)} = \frac{1}{z-2} [1 - (z-2) + (z-2)^2 - (z-2)^3 + \cdots] = \frac{1}{z-2} - 1 + (z-2) - (z-2)^2 + \cdots$$

$$\begin{aligned} 17. f(z) &= \frac{1/3}{z+1} + \frac{2/3}{z-2} = \frac{1}{3(z+1)} + \frac{2}{3} \cdot \frac{1}{-3+(z+1)} = \frac{1}{3(z+1)} - \frac{2}{9} \cdot \frac{1}{1-\frac{z+1}{3}} \\ &= \frac{1}{3(z+1)} - \frac{2}{9} \left[1 + \frac{z+1}{3} + \frac{(z+1)^2}{3^2} + \frac{(z+1)^3}{3^3} + \cdots \right] \\ &= \frac{1}{3(z+1)} - \frac{2}{9} - \frac{2(z+1)}{3^3} - \frac{2(z+1)^2}{3^4} - \cdots \end{aligned}$$

$$\begin{aligned} 18. f(z) &= \frac{1}{3(z+1)} + \frac{2}{3} \cdot \frac{1}{(z+1)-3} = \frac{1}{3(z+1)} + \frac{2}{3(z+1)} \cdot \frac{1}{1-\frac{3}{z+1}} \\ &= \frac{1}{3(z+1)} + \frac{2}{3(z+1)} \left(1 + \frac{3}{z+1} + \frac{3^2}{(z+1)^2} + \frac{3^3}{(z+1)^3} + \cdots \right) \\ &= \frac{1}{z+1} + \frac{2}{(z+1)^2} + \frac{2 \cdot 3}{(z+1)^3} + \frac{2 \cdot 3^2}{(z+1)^4} + \cdots \end{aligned}$$

$$\begin{aligned} 19. f(z) &= \frac{1/3}{z+1} + \frac{2/3}{z-2} = \frac{1}{3z} \cdot \frac{1}{1+\frac{1}{z}} - \frac{1}{3} \cdot \frac{1}{1-\frac{z}{2}} = \frac{1}{3z} \left(1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \cdots \right) - \frac{1}{3} \left(1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} + \cdots \right) \\ &= \cdots - \frac{1}{3z^2} + \frac{1}{3z} - \frac{1}{3} - \frac{z}{3 \cdot 2} - \frac{z^2}{3 \cdot 2^2} - \cdots \end{aligned}$$

$$\begin{aligned} 20. f(z) &= \frac{2/3}{z-2} + \frac{1}{3} \frac{1}{3+(z-2)} = \frac{2/3}{z-2} + \frac{1}{9} \cdot \frac{1}{1+\frac{z-2}{3}} = \frac{2/3}{z-2} + \frac{1}{9} \left(1 + \frac{z-2}{3} + \frac{(z-2)^2}{3^2} + \frac{(z-2)^3}{3^3} + \cdots \right) \\ &= \frac{2}{3(z-2)} + \frac{1}{9} + \frac{z-2}{3^3} + \frac{(z-2)^2}{3^4} + \cdots \end{aligned}$$

$$21. f(z) = \frac{1}{z}(1-z)^{-2} = \frac{1}{z} \left(1 + (-2)(-z) + \frac{(-2)(-3)}{2!}(-z)^2 + \frac{(-2)(-3)(-4)}{3!}(-z)^3 + \cdots \right) = \frac{1}{z} + 2 + 3z + 4z^2 + \cdots$$

$$\begin{aligned} 22. f(z) &= \frac{1}{z^3(1-\frac{1}{z})^2} = \frac{1}{z^3} \left(1 - \frac{1}{z} \right)^{-2} \\ &= \frac{1}{z^3} \left(1 + (-2) \left(-\frac{1}{z} \right) + \frac{(-2)(-3)}{2!} \left(-\frac{1}{z} \right)^2 + \frac{(-2)(-3)(-4)}{3!} \left(-\frac{1}{z} \right)^3 + \cdots \right) \\ &= \frac{1}{z^3} + \frac{2}{z^4} + \frac{3}{z^5} + \frac{4}{z^6} + \cdots \end{aligned}$$

$$\begin{aligned} 23. f(z) &= \frac{1}{(z-2)[1+(z-2)]^3} = \frac{1}{z-2} [1+(z-2)]^{-3} \\ &= \frac{1}{z-2} \left(1 + (-3)(z-2) + \frac{(-3)(-4)}{2!}(z-2)^2 + \frac{(-3)(-4)(-5)}{3!}(z-2)^3 + \cdots \right) \\ &= \frac{1}{z-2} - 3 + 6(z-2) - 10(z-2)^2 + \cdots \end{aligned}$$

24. $f(z) = \frac{1}{(z-3)^3} \cdot \frac{-1}{1-(z-1)} = \frac{-1}{(z-1)^3} [1 + (z-1) + (z-1)^2 + (z-1)^3 + \cdots]$
 $= -\frac{1}{(z-1)^3} - \frac{1}{(z-1)^2} - \frac{1}{z-1} - 1 - (z-1) - \cdots$
25. $f(z) = \frac{3}{z} + \frac{4}{z-1} = \frac{3}{z} - 4 \cdot \frac{1}{1-z} = \frac{3}{z} - 4(1 + z + z^2 + z^3 + \cdots) = \frac{3}{z} - 4 - 4z - 4z^2 - \cdots$
26. $f(z) = \frac{4}{z-1} + 3 \cdot \frac{1}{1+(z-1)} = \frac{4}{z-1} + 3(1 - (z-1) + (z-1)^2 - (z-1)^3 + \cdots) = \frac{4}{z-1} + 3 - 3(z-1) + 3(z-1)^2 - \cdots$
27. $f(z) = z + \frac{2}{z-2} = 1 + (z-1) + \frac{2}{-1+z-1} = 1 + (z-1) + \frac{2}{z-1} \cdot \frac{1}{1 - \frac{1}{z-1}}$
 $= 1 + (z-1) + \frac{2}{z-1} \left(1 + \frac{1}{z-1} + \frac{1}{(z-1)^2} + \frac{1}{(z-1)^3} + \cdots \right) = \cdots + \frac{2}{(z-1)^2} + \frac{2}{z-1} + 1 + (z-1)$
28. $f(z) = z + \frac{2}{z-2} = \frac{2}{z-2} + 2 + (z-2)$

EXERCISES 19.4

Zeros and Poles

1. Using $e^{2z} = \sum_{k=0}^{\infty} \frac{2^k z^k}{k!}$ we obtain

$$\frac{e^{2z} - 1}{z} = \frac{\left(1 + \frac{2}{1!}z + \frac{2^2}{2!}z^2 + \frac{2^3}{3!}z^3 + \cdots \right) - 1}{z} = \frac{1}{z} \left(\frac{2}{1!}z + \frac{2^2}{2!}z^2 + \frac{2^3}{3!}z^3 + \cdots \right) = \frac{2}{1!} + \frac{2^2}{2!}z + \frac{2^3}{3!}z^2 + \cdots$$

From the form of the last series we see that $z = 0$ is a removable singularity. Define $f(0) = 2$.

2. Using $\sin 4z = \sum_{k=0}^{\infty} (-1)^k \frac{(4z)^{2k+1}}{(2k+1)!}$ we obtain

$$\begin{aligned} \frac{\sin 4z - 4z}{z^2} &= \frac{\left(\frac{4}{1!}z - \frac{4^3}{3!}z^3 + \frac{4^5}{5!}z^5 - \frac{4^7}{7!}z^7 + \cdots \right) - 4z}{z^2} = \frac{1}{z^2} \left(-\frac{4^3}{3!}z^3 + \frac{4^5}{5!}z^5 - \frac{4^7}{7!}z^7 + \cdots \right) \\ &= -\frac{4^3}{3!}z + \frac{4^5}{5!}z^3 - \frac{4^7}{7!}z^5 + \cdots \end{aligned}$$

From the form of the last series we see that $z = 0$ is a removable singularity. Define $f(0) = 0$.

3. Since $f(-2+i) = f'(-2+i) = 0$ and $f''(z) = 2$ for all z , $z = -2+i$ is a zero of order two.
4. Write $f(z) = z^4 - 16 = (z^2 - 4)(z^2 + 4) = (z-2)(z+2)(z-2i)(z+2i)$ to see that 2, -2, 2i, and -2i are zeros of f . Now $f'(z) = 4z^3$ and $f'(2) \neq 0$, $f'(-2) \neq 0$, $f'(2i) \neq 0$, and $f'(-2i) \neq 0$. This indicates that each zero is of order one.
5. Write $f(z) = z^2(z^2 + 1) = z^2(z-i)(z+i)$ to see that 0, i , and $-i$ are zeros of f . Now $f'(z) = 4z^3 + 2z$ and $f'(i) \neq 0$ and $f'(-i) \neq 0$. This indicates that $z = i$ and $z = -i$ are zeros of order one. However $f'(0) = 0$, but $f''(0) = 2 \neq 0$. Hence $z = 0$ is a zero of order two.

19.4 Zeros and Poles

6. Write $f(z) = (z^2 + 9)/z = (z - 3i)(z + 3i)/z$ to see that $3i$ and $-3i$ are zeros of f . Now $f'(z) = 1 - 9/z^2$ and $f'(3i) = f'(-3i) = 2 \neq 0$. This indicates that each zero is of order one.
7. Write $f(z) = e^z(e^z - 1)$ to see that $2n\pi i$, $n = 0, \pm 1, \pm 2, \dots$ are zeros of f . Now $f'(z) = 2e^{2z} - e^z$ and $f'(2n\pi i) = 2e^{4n\pi i} - e^{2n\pi i} = 1 \neq 0$. This indicates that each zero is of order one.
8. The zeros of f are the zeros of $\sin z$, that is, $n\pi$, $n = 0, \pm 1, \pm 2, \dots$. From $f'(z) = 2 \sin z \cos z$ we see $f'(n\pi) = 0$. From $f''(z) = 2(-\sin^2 z + \cos^2 z)$ we see $f''(n\pi) \neq 0$. This indicates that each zero is of order two.

9. From
$$f(z) = z(1 - \cos z^2) = z \left(-\frac{z^4}{2!} + \frac{z^8}{4!} - \dots \right) = z^5 \left(-\frac{1}{2!} + \frac{z^4}{4!} - \dots \right)$$

we see that $z = 0$ is a zero of order five.

10. From
$$f(z) = z - \sin z = \frac{z^3}{3!} - \frac{z^5}{5!} + \dots = z^3 \left(\frac{1}{3!} - \frac{z^2}{5!} + \dots \right)$$

we see that $z = 0$ is a zero of order three.

11. From
$$f(z) = 1 - e^{z-1} = -\frac{z-1}{1!} - \frac{(z-1)^2}{2!} - \dots = (z-1) \left(1 - \frac{z-1}{2!} - \dots \right)$$

we see that $z = 1$ is a zero of order one.

12. From the series $e^z = -\sum_{k=0}^{\infty} \frac{(z - \pi i)^k}{k!}$ centered at πi and

$$\begin{aligned} f(z) &= 1 - \pi i + z + e^z = 1 - \pi i + z + \left(-1 - \frac{z - \pi i}{1!} - \frac{(z - \pi i)^2}{2!} - \frac{(z - \pi i)^3}{3!} - \dots \right) \\ &= -\frac{(z - \pi i)^2}{2!} - \frac{(z - \pi i)^3}{3!} - \dots = (z - \pi i)^2 \left(-\frac{1}{2!} - \frac{z - \pi i}{3!} - \dots \right) \end{aligned}$$

we see that $z = \pi i$ is a zero of order two.

13. From
$$f(z) = \frac{3z - 1}{[(z - (-1 + 2i)][z - (-1 - 2i)]}$$

and Theorem 19.11 we see that $-1 + 2i$ and $-1 - 2i$ are simple poles.

14. From $f(z) = \frac{5z^2 - 6}{z^2}$ and Theorem 19.11 we see that 0 is a pole of order two.

15. From $f(z) = \frac{1 + 4i}{(z + 2)(z + i)^4}$ and Theorem 19.11 we see that -2 is a simple pole and $-i$ is a pole of order four.

16. From
$$f(z) = \frac{z - 1}{(z + 1)^2 \left[z - \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) \right] \left[z - \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) \right]}$$

and Theorem 19.11 we see that -1 is a pole of order two and $\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $\frac{1}{2} - \frac{\sqrt{3}}{2}i$ are simple poles.

17. Since $\sin z$ and $\cos z$ are analytic at $n\pi$, $n = 0, \pm 1, \pm 2, \dots$, $\sin z$ has zeros of order one at $n\pi$, and $\cos n\pi \neq 0$, it follows from Theorem 19.11 that the numbers $n\pi$, $n = 0, \pm 1, \pm 2, \dots$ are simple poles of $f(z) = \tan z$.

18. From $z^2 \sin \pi z = z^3 \left(\pi - \frac{\pi^3 z^2}{3!} + \dots \right)$ we see $z = 0$ is a zero of order three. From $f(z) = \frac{\cos \pi z}{z^2 \sin \pi z}$ and Theorem 19.11 we see 0 is a pole of order three. The numbers $n, n \pm 1, \pm 2, \dots$ are simple poles.

19. From the Laurent series

$$f(z) = \frac{1 - \cosh z}{z^4} = \frac{1 - \left(1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \frac{z^6}{6!} + \dots \right)}{z^4} = -\frac{1}{2!z^2} - \frac{1}{4!} - \frac{z^2}{6!} - \dots$$

we see that 0 is a pole of order two.

20. From the Laurent series

$$f(z) = \frac{e^z}{z^2} = \frac{\left(1 + \frac{z}{1!} + \frac{z^2}{2!} + \cdots\right)}{z^2} = \frac{1}{z^2} + \frac{1}{z} + \frac{1}{2!} + \cdots$$

we see that 0 is a pole of order two.

21. From $1 - e^z = 1 - \left(1 + \frac{z}{1!} + \frac{z^2}{2!} + \cdots\right) = z\left(-1 - \frac{z}{2!} - \cdots\right)$ we see that $z = 0$ is a zero of order one. By periodicity of e^z it follows that $z = 2n\pi i$, $n = 0, \pm 1, \pm 2, \dots$ are zeros of order one. From $f(z) = \frac{1}{1 - e^z}$ and Theorem 19.11 we see that the numbers $2n\pi i$, $n = 0, \pm 1, \pm 2, \dots$ are simple poles.

22. $z = 0$ is a removable singularity of the function $(\sin z)/z$. From $f(z) = \frac{\sin z}{z(z-1)}$ we see that only 1 is a (simple) pole.

23. The function $f(z) = \frac{\sin(1/z)}{\cos(1/z)}$ fails to be defined at $z = 0$ and at the solutions of $\cos \frac{1}{z} = 0$, that is, at $\frac{1}{z} = (2n+1)\frac{\pi}{2}$, $n = 0, \pm 1, \pm 2, \dots$. Since $z = \frac{2}{(2n+1)\pi}$, $n = 0, \pm 1, \pm 2, \dots$ we see that in any neighborhood of $z = 0$ there are points at which f is not defined and thus not analytic. Hence $z = 0$ is a non-isolated singularity.

24. From the Laurent series

$$f(z) = z^3 \left[\frac{1}{z} - \frac{1}{3!} \left(\frac{1}{z}\right)^3 + \frac{1}{5!} \left(\frac{1}{z}\right)^5 - \frac{1}{7!} \left(\frac{1}{z}\right)^7 + \cdots \right] = z^2 - \frac{1}{3!} + \frac{1}{5!z^2} - \frac{1}{7!z^4} + \cdots, \quad 0 < |z|,$$

we see that the principal part contains an infinite number of nonzero terms. Hence $z = 0$ is an essential singularity.

EXERCISES 19.5

Residues and Residue Theorem

$$\begin{aligned} 1. \quad f(z) &= \frac{2}{5(z-1)} \cdot \frac{1}{1 + \frac{z-1}{5}} = \frac{2}{5(z-1)} \left(1 - \frac{z-1}{5} + \frac{(z-1)^2}{5^2} - \frac{(z-1)^3}{5^3} + \cdots \right) \\ &= \frac{2/5}{z-1} - \frac{2}{25} + \frac{2(z-1)}{5^3} - \frac{2(z-1)^2}{5^4} + \cdots \end{aligned}$$

$$\text{Res}(f(z), 1) = 2/5$$

$$\begin{aligned} 2. \quad f(z) &= \frac{1}{z^3}(1-z)^{-3} = \frac{1}{z^3} \left(1 + (-3)(-z) + \frac{(-3)(-4)}{2!}(-z)^2 + \frac{(-3)(-4)(-5)}{3!}(-z)^3 + \cdots \right) \\ &= \frac{1}{z^3} + \frac{3}{z^2} + \frac{6}{z} + 10 + \cdots \end{aligned}$$

$$\text{Res}(f(z), 0) = 6$$

$$3. \quad f(z) = -\frac{3}{z} - \frac{1}{z-2} = -\frac{3}{z} + \frac{1}{2} \cdot \frac{1}{1 - \frac{z}{2}} = -\frac{3}{z} + \frac{1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{2^2} + \frac{z^3}{2^3} + \cdots \right) = -\frac{3}{z} + \frac{1}{2} + \frac{z}{2^2} + \frac{z^2}{2^3} + \cdots$$

19.5 Residues and Residue Theorem

$$\operatorname{Res}(f(z), 0) = -3$$

$$4. f(z) = (z+3)^2 \left(\frac{2}{z+3} - \frac{2^3}{3!(z+3)^3} + \frac{2^5}{5!(z+3)^5} + \cdots \right) = \cdots + \frac{2^5}{5!(z+3)^3} - \frac{2^3}{3!(z+3)} + 2(z+3)$$

$$\operatorname{Res}(f(z), -3) = -\frac{4}{3}$$

$$5. f(z) = e^{-2/z^2} = \sum_{k=0}^{\infty} \frac{(-2/z^2)^k}{k!} = \cdots - \frac{2^3}{3!z^6} + \frac{2^2}{2!z^4} - \frac{2}{1!z^2} + 1; \quad \operatorname{Res}(f(z), 0) = 0$$

$$6. f(z) = \frac{e^{-2}}{(z-2)^2} e^{-(z-2)} = \frac{e^{-2}}{(z-2)^2} \left(1 - \frac{z-2}{1!} + \frac{(z-2)^2}{2!} - \frac{(z-2)^3}{3!} + \cdots \right)$$

$$= \frac{e^{-2}}{(z-2)^2} - \frac{e^{-2}}{z-2} + \frac{e^{-2}}{2} - \frac{e^{-2}(z-2)}{3!} + \cdots$$

$$\operatorname{Res}(f(z), 2) = -e^{-2}$$

$$7. \operatorname{Res}(f(z), 4i) = \lim_{z \rightarrow 4i} (z-4i) \cdot \frac{z}{(z-4i)(z+4i)} = \lim_{z \rightarrow 4i} \frac{z}{z+4i} = \frac{1}{2}$$

$$\operatorname{Res}(f(z), -4i) = \lim_{z \rightarrow -4i} (z+4i) \cdot \frac{z}{(z-4i)(z+4i)} = \lim_{z \rightarrow -4i} \frac{z}{z-4i} = \frac{1}{2}$$

$$8. \operatorname{Res}(f(z), 1/2) = \lim_{z \rightarrow 1/2} (z-1/2) \frac{4z+8}{2(z-1/2)} = \lim_{z \rightarrow 1/2} (2z+4) = 5$$

$$9. \operatorname{Res}(f(z), 1) = \lim_{z \rightarrow 1} (z-1) \frac{1}{z^2(z+2)(z-1)} = \lim_{z \rightarrow 1} \frac{1}{z^2(z+2)} = \frac{1}{3}$$

$$\operatorname{Res}(f(z), -2) = \lim_{z \rightarrow -2} (z+2) \frac{1}{z^2(z+2)(z-1)} = \lim_{z \rightarrow -2} \frac{1}{z^2(z-1)} = -\frac{1}{12}$$

$$\operatorname{Res}(F(z), 0) = \frac{1}{1!} \lim_{z \rightarrow 0} \frac{d}{dz} \left[z^2 \cdot \frac{1}{z^2(z+2)(z-1)} \right] = \lim_{z \rightarrow 0} \frac{-2z-1}{(z+2)^2(z-1)^2} = -\frac{1}{4}$$

$$10. \operatorname{Res}(f(z), 1+i) = \frac{1}{1!} \lim_{z \rightarrow 1+i} \frac{d}{dz} \left[(z-1-i)^2 \cdot \frac{1}{(z-1-i)^2(z-1+i)^2} \right] = \lim_{z \rightarrow 1+i} \frac{-2}{(z-1+i)^3} = -\frac{1}{4}i$$

$$\operatorname{Res}(f(z), 1-i) = \frac{1}{1!} \lim_{z \rightarrow 1-i} \frac{d}{dz} \left[(z-1+i)^2 \cdot \frac{1}{(z-1-i)^2(z-1+i)^2} \right] = \lim_{z \rightarrow 1-i} \frac{-2}{(z-1-i)^3} = \frac{1}{4}i$$

$$11. \operatorname{Res}(f(z), -1) = \lim_{z \rightarrow -1} (z+1) \cdot \frac{5z^2-4z+3}{(z+1)(z+2)(z+3)} = \lim_{z \rightarrow -1} \frac{5z^2-4z+3}{(z+2)(z+3)} = 6$$

$$\operatorname{Res}(f(z), -2) = \lim_{z \rightarrow -2} (z+2) \cdot \frac{5z^2-4z+3}{(z+1)(z+2)(z+3)} = \lim_{z \rightarrow -2} \frac{5z^2-4z+3}{(z+1)(z+3)} = -31$$

$$\operatorname{Res}(f(z), -3) = \lim_{z \rightarrow -3} (z+3) \cdot \frac{5z^2-4z+3}{(z+1)(z+2)(z+3)} = \lim_{z \rightarrow -3} \frac{5z^2-4z+3}{(z+1)(z+2)} = 30$$

$$12. \operatorname{Res}(f(z), -3) = \lim_{z \rightarrow -3} (z+3) \cdot \frac{2z-1}{(z-1)^4(z+3)} = \lim_{z \rightarrow -3} \frac{2z-1}{(z-1)^4} = -\frac{7}{256}$$

$$\operatorname{Res}(f(z), 1) = \frac{1}{3!} \lim_{z \rightarrow 1} \frac{d^3}{dz^3} \left[(z-1)^4 \cdot \frac{2z-1}{(z-1)^4(z+3)} \right] = \frac{1}{6} \lim_{z \rightarrow 1} \frac{-42}{(z+3)^4} = -\frac{7}{256}$$

$$13. \operatorname{Res}(f(z), 0) = \frac{1}{1!} \lim_{z \rightarrow 0} \frac{d}{dz} \left[z^2 \cdot \frac{\cos z}{z^2(z-\pi)^3} \right] = \lim_{z \rightarrow 0} \frac{-(z-\pi) \sin z - 3 \cos z}{(z-\pi)^4} = -\frac{3}{\pi^4}$$

$$\operatorname{Res}(f(z), \pi) = \frac{1}{2!} \lim_{z \rightarrow \pi} \frac{d^2}{dz^2} \left[(z-\pi)^3 \cdot \frac{\cos z}{z^2(z-\pi)^3} \right] = \frac{1}{2} \lim_{z \rightarrow \pi} \frac{-z^2 \cos z + 4z \sin z + 6 \cos z}{z^4} = \frac{\pi^2 - 6}{2\pi^4}$$

14. Using $\frac{d}{dz}(e^z - 1) = e^z$ and the result in (4) in the text,

$$\operatorname{Res}(f(z), 2n\pi i) = \frac{e^z}{e^z} \Big|_{z=2n\pi i} = 1.$$

15. Using $\frac{d}{dz} \cos z = -\sin z$ and the result in (4) in the text,

$$\operatorname{Res}\left(f(z), (2n+1)\frac{\pi}{2}\right) = \frac{1}{-\sin z} \Big|_{z=(2n+1)\frac{\pi}{2}} = \frac{1}{-\sin(2n+1)\frac{\pi}{2}} = (-1)^{n+1}.$$

16. $z = 0$ is a pole of order two. Thus by (2) in the text and L'Hôpital's rule,

$$\operatorname{Res}(f(z), 0) = \frac{1}{1!} \lim_{z \rightarrow 0} \frac{d}{dz} \left[z^2 \cdot \frac{1}{z \sin z} \right] = \lim_{z \rightarrow 0} \frac{\sin z - z \cos z}{\sin^2 z} = \lim_{z \rightarrow 0} \frac{\cos z + z \sin z - \cos z}{2 \sin z \cos z} = \lim_{z \rightarrow 0} \frac{z}{2 \cos z} = 0.$$

For the simple poles at $z = n\pi$, $n = \pm 1, \pm 2, \dots$ we have from (4) in the text,

$$\operatorname{Res}(f(z), n\pi) = \frac{1}{z \cos z + \sin z} \Big|_{z=n\pi} = \frac{(-1)^n}{n\pi}.$$

17. (a) $\oint_C \frac{1}{(z-1)(z+2)^2} dz = 0$ by Theorem 18.4.

(b) $\oint_C \frac{1}{(z-1)(z+2)^2} dz = 2\pi i \operatorname{Res}(f(z), 1) = \frac{2\pi}{9} i$

(c) $\oint_C \frac{1}{(z-1)(z+2)^2} dz = 2\pi i [\operatorname{Res}(f(z), 1) + \operatorname{Res}(f(z), -2)] = 2\pi i \left[\frac{1}{9} + \left(-\frac{1}{9} \right) \right] = 0$

18. (a) $\oint_C \frac{z+1}{z^2(z-2i)} dz = 2\pi i \operatorname{Res}(f(z), 0) = \pi \left(-1 + \frac{1}{2} i \right)$

(b) $\oint_C \frac{z+1}{z^2(z-2i)} dz = 2\pi i \operatorname{Res}(f(z), 2i) = \pi \left(1 - \frac{1}{2} i \right)$

(c) $\oint_C \frac{z+1}{z^2(z-2i)} dz = 2\pi i [\operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), 2i)] = 2\pi i \left[\frac{1}{4} + \frac{1}{2} i + \left(-\frac{1}{4} - \frac{1}{2} i \right) \right] = 0$

19. (a) From the Laurent series $z^3 e^{-1/z^2} = \dots - \frac{1}{3!z^3} + \frac{1}{2!z} - z + z^3$ we see $\operatorname{Res}(f(z), 0) = 1/2$. Hence

$$\oint_C z^3 e^{-1/z^2} dz = 2\pi i \operatorname{Res}(f(z), 0) = \pi i.$$

(b) $\oint_C z^3 e^{-1/z^2} dz = 2\pi i \operatorname{Res}(f(z), 0) = \pi i$

(c) $\oint_C z^3 e^{1/z^2} dz = 0$ by Theorem 18.4.

20. (a) $\oint_C \frac{1}{z \sin z} dz = 0$ by Theorem 18.4.

- (b) $z = 0$ is a pole of order two (see Problem 16). Thus

$$\oint_C \frac{1}{z \sin z} dz = 2\pi i \operatorname{Res}(f(z), 0) = 2\pi i(0) = 0.$$

(c) $\oint_C \frac{1}{z \sin z} dz = 2\pi i [\operatorname{Res}(f(z), -\pi) + \operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), \pi)] = 2\pi i \left[\frac{1}{\pi} + 0 + \left(-\frac{1}{\pi} \right) \right] = 0$

19.5 Residues and Residue Theorem

21. $\oint_C \frac{1}{z^2 + 4z + 13} dz = 2\pi i \operatorname{Res}(f(z), -2 + 3i) = \frac{\pi}{3}$
22. $\oint_C 1z^3(z-1)^4 dz = 2\pi i \operatorname{Res}(f(z), 1) = -20\pi i$
23. $\oint_C \frac{z}{z^4 - 1} dz = 2\pi i [\operatorname{Res}(f(z), -1) + \operatorname{Res}(f(z), 1) + \operatorname{Res}(f(z), -i) + \operatorname{Res}(f(z), i)] = 2\pi i \left[\frac{1}{4} + \frac{1}{4} - \frac{1}{4} - \frac{1}{4} \right] = 0$
24. $\oint_C \frac{z}{(z+1)(z^2+1)} dz = 2\pi i [\operatorname{Res}(f(z), i) + \operatorname{Res}(f(z), -i)] = 2\pi i \left[\frac{1}{4} - \frac{1}{4}i + \frac{1}{4} + \frac{1}{4}i \right] = \pi i$
25. $\oint_C \frac{ze^z}{z^2 - 1} dz = 2\pi i [\operatorname{Res}(f(z), 1) + \operatorname{Res}(f(z), -1)] = 2\pi i \left[\frac{e}{2} + \frac{e^{-1}}{2} \right] = 2\pi i \cosh 1$
26. $\oint_C \frac{e^z}{z^3 + 2z^2} dz = 2\pi i [\operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), -2)] = 2\pi i \left[\frac{1}{2} + \frac{e^{-2}}{4} \right] = \pi i \left(1 + \frac{1}{2}e^{-2} \right)$
27. $\oint_C \frac{\tan z}{z} dz = 2\pi i \operatorname{Res} \left(f(z), \frac{\pi}{2} \right) = -4i$. Note: $z = 0$ is not a pole. See Example 1, Section 19.4.
28. $\oint_C \frac{\cot \pi z}{z^2} dz = 2\pi i \operatorname{Res}(f(z), 0) = 2\pi i \left(-\frac{\pi}{3} \right) = -\frac{2\pi^2}{3} i$

Note: $z = 0$ is a pole of order three. Use L'Hôpital's rule (or *Mathematica*) to show that

$$\operatorname{Res}(f(z), 0) = \frac{1}{2} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} z \cot \pi z = \frac{1}{2} \lim_{z \rightarrow 0} [-2\pi \csc^2 \pi z + 2\pi^2 z \cot \pi z \csc^2 \pi z] = \frac{1}{2} \left(-\frac{2\pi}{3} \right) = -\frac{\pi}{3}.$$

29. $\oint_C \cot \pi z dz = 2\pi i [\operatorname{Res}(f(z), 1) + \operatorname{Res}(f(z), 2) + \operatorname{Res}(f(z), 3)] = 2\pi i \left[\frac{1}{\pi} + \frac{1}{\pi} + \frac{1}{\pi} \right] = 6i$
30. $\oint_C \frac{2z-1}{z^2(z^3+1)} dz = 2\pi i \left[\operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), -1) + \operatorname{Res} \left(f(z), \frac{1}{2} + \frac{\sqrt{3}}{2}i \right) \right]$
 $= 2\pi i \left[2 + (-1) + \left(-\frac{1}{2} - \frac{1}{6}\sqrt{3}i \right) \right] = \pi \left(\frac{\sqrt{3}}{3} + i \right)$
31. $\oint_C \frac{e^{iz} + \sin z}{(z-\pi)^4} dz = 2\pi i \operatorname{Res}(f(z), \pi) = \pi \left(-\frac{1}{3} + \frac{1}{3}i \right)$
32. $\oint_C \frac{\cos z}{(z-1)^2(z^2+9)} dz = 2\pi i \operatorname{Res}(f(z), 1) = 2\pi i (-0.02 \cos 1 - 0.1 \sin 1) = -0.5966i$

EXERCISES 19.6

Evaluation of Real Integrals

1. $\int_0^{2\pi} \frac{d\theta}{1 + \frac{1}{2} \sin \theta} = \oint_C \frac{4}{z^2 + 4iz - 1} dz = (4)2\pi i \operatorname{Res}(f(z), (\sqrt{3} - 2)i) = \frac{4\pi}{\sqrt{3}}$
2. $\int_0^{2\pi} \frac{d\theta}{10 - 6 \cos \theta} = \frac{1}{2} \cdot \left(\frac{-2}{i} \right) \oint_C \frac{dz}{(3z-1)(z-3)} = (i)2\pi i \operatorname{Res} \left(f(z), \frac{1}{3} \right) = \frac{\pi}{4}$

3. $\int_0^{2\pi} \frac{\cos \theta}{3 + \sin \theta} d\theta = \oint_C \frac{z^2 + 1}{z(z^2 + 6iz - 1)} dz = 2\pi i [\text{Res}(f(z), 0) + \text{Res}(f(z), -3 + 2\sqrt{2}i)] = 0$
4. $\int_0^{2\pi} \frac{1}{1 + 3\cos^2 \theta} d\theta = \frac{4}{i} \oint_C \frac{z}{3z^4 + 10z^2 + 3} dz = \left(\frac{4}{i}\right) 2\pi i \left[\text{Res}\left(f(z), \left(\frac{\sqrt{3}}{3}i\right)\right) + \text{Res}\left(f(z), -\frac{\sqrt{3}}{3}i\right) \right] = \pi$
5. $\int_0^\pi \frac{d\theta}{2 - \cos \theta} = \frac{1}{2} \int_0^{2\pi} \frac{d\theta}{2 - \cos \theta} = -\frac{1}{i} \oint_C \frac{dz}{z^2 - 4z + 1} = \left(-\frac{1}{i}\right) 2\pi i \text{Res}(f(z), 2 - \sqrt{3}) = \frac{\pi}{\sqrt{3}}$
6. $\int_0^\pi \frac{d\theta}{1 + \sin^2 \theta} = \frac{1}{2} \int_0^{2\pi} \frac{d\theta}{1 + \sin^2 \theta} = -\frac{2}{i} \oint_C \frac{z}{z^4 - 6z^2 + 1} dz$
 $= \left(-\frac{2}{i}\right) 2\pi i [\text{Res}(f(z), \sqrt{3 - 2\sqrt{2}}) + \text{Res}(f(z), -\sqrt{3 - 2\sqrt{2}})] = \frac{\pi}{\sqrt{2}}$
7. $\int_0^{2\pi} \frac{\sin^2 \theta}{5 + 4\cos \theta} d\theta = -\frac{1}{4i} \oint_C \frac{(z^2 - 1)^2}{z^2(2z^2 + 5z + 2)} dz = \left(-\frac{1}{4i}\right) 2\pi i \left[\text{Res}(f(z), 0) + \text{Res}\left(f(z), -\frac{1}{2}\right) \right] = \frac{\pi}{4}$
8. $\int_0^{2\pi} \frac{\cos^2 \theta}{3 - \sin \theta} d\theta = \frac{1}{2i} \oint_C \frac{z^4 + 2z^2 + 1}{z^2(iz^2 + 6z - i)} dz = \left(\frac{1}{2i}\right) 2\pi i [\text{Res}(f(z), 0) + \text{Res}(f(z), 3 - 2\sqrt{2}i)] = \pi[6 - 4\sqrt{2}]$
9. We use $\cos 2\theta = (z^2 + z^{-2})/2$.
 $\int_0^{2\pi} \frac{\cos 2\theta}{5 - 4\cos \theta} d\theta = \frac{i}{2} \oint_C \frac{z^4 + 1}{z^2(2z^2 - 5z + 2)} dz = \left(\frac{i}{2}\right) 2\pi i \left[\text{Res}(f(z), 0) + \text{Res}\left(f(z), \frac{1}{2}\right) \right] = \frac{\pi}{6}$
10. $\int_0^{2\pi} \frac{1}{\cos \theta + 2\sin \theta + 3} d\theta = \frac{2}{i} \oint_C \frac{1}{(1 - 2i)z^2 + 6z + 1 + 2i} = \left(\frac{2}{i}\right) 2\pi i \text{Res}\left(f(z), -\frac{1}{5} - \frac{2}{5}i\right) = \pi$
11. $\int_{-\infty}^{\infty} \frac{1}{x^2 - 2x + 2} dx = 2\pi i \text{Res}(f(z), 1 + i) = \pi$
12. $\int_{-\infty}^{\infty} \frac{1}{x^2 - 2x + 25} dx = 2\pi i \text{Res}(f(z), 1 + 2\sqrt{6}i) = \frac{\pi}{2\sqrt{6}}$
13. $\int_{-\infty}^{\infty} \frac{1}{(x^2 + 4)^2} dx = 2\pi i \text{Res}(f(z), 2i) = \frac{\pi}{16}$
14. $\int_{-\infty}^{\infty} \frac{x^2}{(x^2 + 1)^2} dx = 2\pi i \text{Res}(f(z), i) = \frac{\pi}{2}$
15. $\int_{-\infty}^{\infty} \frac{1}{(x^2 + 1)^3} dx = 2\pi i \text{Res}(f(z), i) = \frac{3\pi}{8}$
16. $\int_{-\infty}^{\infty} \frac{x}{(x^2 + 4)^3} dx = 0$ (The integrand is an odd function)
17. $\int_{-\infty}^{\infty} \frac{2x^2 - 1}{x^4 + 5x^2 + 4} dx = 2\pi i [\text{Res}(f(z), i) + \text{Res}(f(z), 2i)] = \frac{\pi}{2}$
18. $\int_{-\infty}^{\infty} \frac{dx}{(x^2 + 1)^2(x^2 + 9)} = 2\pi i [\text{Res}(f(z), i) + \text{Res}(f(z), 3i)] = \frac{5\pi}{96}$
19. $\int_0^\infty \frac{x^2 + 1}{x^4 + 1} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{x^2 + 1}{x^4 + 1} dx = \pi i \text{Res}(f(z), i) = \frac{\pi}{\sqrt{2}}$
20. $\int_0^\infty \frac{1}{x^6 + 1} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{1}{x^6 + 1} dx = \pi i \left[\text{Res}\left(f(z), \frac{\sqrt{3}}{2} + \frac{1}{2}i\right) + \text{Res}(f(z), i) + \text{Res}\left(f(z), \frac{-\sqrt{3}}{2} + \frac{1}{2}i\right) \right] = \frac{\pi}{3}$

19.6 Evaluation of Real Integrals

21. $\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2+1} dx = 2\pi i \operatorname{Res}(f(z), i) = \pi e^{-1}$. Therefore, $\int_{-\infty}^{\infty} \frac{\cos x}{x^2+1} dx = \operatorname{Re} \left(\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2+1} dx \right) = \pi e^{-1}$.

22. $\int_{-\infty}^{\infty} \frac{e^{2ix}}{x^2+1} dx = 2\pi i \operatorname{Res}(f(z), i) = \pi e^{-2}$. Therefore, $\int_{-\infty}^{\infty} \frac{\cos 2x}{x^2+1} dx = \operatorname{Re} \left(\int_{-\infty}^{\infty} \frac{e^{2ix}}{x^2+1} dx \right) = \pi e^{-2}$.

23. $\int_{-\infty}^{\infty} \frac{x e^{ix}}{x^2+1} dx = 2\pi i \operatorname{Res}(f(z), i) = \pi e^{-1} i$. Therefore, $\int_{-\infty}^{\infty} \frac{x \sin x}{x^2+1} dx = \operatorname{Im} \left(\int_{-\infty}^{\infty} \frac{x e^{ix}}{x^2+1} dx \right) = \pi e^{-1}$.

24. $\int_{-\infty}^{\infty} \frac{e^{ix}}{(x^2+4)^2} dx = 2\pi i \operatorname{Res}(f(z), 2i) = \frac{3e^{-2}}{16} \pi$; $\int_{-\infty}^{\infty} \frac{\cos x}{(x^2+4)^2} dx = \operatorname{Re} \left(\int_{-\infty}^{\infty} \frac{e^{ix}}{(x^2+4)^2} dx \right) = \frac{3e^{-2}}{16} \pi$.
Therefore,

$$\int_0^{\infty} \frac{\cos x}{(x^2+4)^2} dx = \frac{1}{2} \left(\frac{3e^{-2}}{16} \pi \right) = \frac{3e^{-2}}{32} \pi.$$

25. $\int_{-\infty}^{\infty} \frac{e^{3ix}}{(x^2+1)^2} dx = 2\pi i \operatorname{Res}(f(z), i) = 2\pi e^{-3}$; $\int_{-\infty}^{\infty} \frac{\cos 3x}{(x^2+1)^2} dx = \operatorname{Re} \left(\int_{-\infty}^{\infty} \frac{e^{3ix}}{(x^2+1)^2} dx \right) = 2\pi e^{-3}$.
Therefore,

$$\int_0^{\infty} \frac{\cos 3x}{(x^2+1)^2} dx = \frac{1}{2} (2\pi e^{-3}) = \pi e^{-3}.$$

26. $\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2+4x+5} dx = 2\pi i \operatorname{Res}(f(z), -2+i) = \pi e^{-1-2i}$. Therefore

$$\int_{-\infty}^{\infty} \frac{\sin x}{x^2+4x+5} dx = \operatorname{Im} \left(\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2+4x+5} dx \right) = -\pi e^{-1} \sin 2$$

27. $\int_{-\infty}^{\infty} \frac{e^{2ix}}{x^4+1} dx = 2\pi i \left[\operatorname{Res} \left(f(z), \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) + \operatorname{Res} \left(f(z), -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) \right]$
 $= 2\pi i \left[\left(-\frac{\sqrt{2}}{8} - \frac{\sqrt{2}}{8} i \right) e^{(-\sqrt{2}+\sqrt{2}i)} + \left(\frac{\sqrt{2}}{8} - \frac{\sqrt{2}}{8} i \right) e^{(-\sqrt{2}-\sqrt{2}i)} \right]$
 $= \pi e^{-\sqrt{2}} \left[\frac{\sqrt{2}}{2} \cos \sqrt{2} + \frac{\sqrt{2}}{2} \sin \sqrt{2} \right]$

$$\int_{-\infty}^{\infty} \frac{\cos 2x}{x^4+1} dx = \operatorname{Re} \left(\int_{-\infty}^{\infty} \frac{e^{2ix}}{x^4+1} dx \right) = \pi e^{-\sqrt{2}} \left[\frac{\sqrt{2}}{2} \cos \sqrt{2} + \frac{\sqrt{2}}{2} \sin \sqrt{2} \right]$$

Therefore

$$\int_0^{\infty} \frac{\cos 2x}{x^4+1} dx = \pi e^{-\sqrt{2}} \frac{\sqrt{2}}{4} (\cos \sqrt{2} + \sin \sqrt{2}).$$

28. $\int_{-\infty}^{\infty} \frac{x e^{ix}}{x^4+1} dx = 2\pi i \left[\operatorname{Res} \left(f(z), \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) + \operatorname{Res} \left(f(z), -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) \right]$
 $= 2\pi i \left[-\frac{i}{4} e^{(-1/\sqrt{2}+i/\sqrt{2})} + \frac{i}{4} e^{(-1/\sqrt{2}-i/\sqrt{2})} \right] = \left(\pi e^{-1/\sqrt{2}} \sin \frac{1}{\sqrt{2}} \right) i$

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^4+1} dx = \operatorname{Im} \left(\int_{-\infty}^{\infty} \frac{x e^{ix}}{x^4+1} dx \right) = \pi e^{-1/\sqrt{2}} \sin \frac{1}{\sqrt{2}}$$

Therefore

$$\int_0^{\infty} \frac{x \sin x}{x^4+1} dx = \frac{\pi}{2} e^{-1/\sqrt{2}} \sin \frac{1}{\sqrt{2}}.$$

$$29. \int_{-\infty}^{\infty} \frac{e^{ix}}{(x^2+1)(x^2+9)} dx = 2\pi i [\text{Res}(f(z), i) + \text{Res}(f(z), 3i)] = 2\pi i \left[-\frac{i}{16} e^{-1} + \frac{i}{48} e^{-3} \right] = \frac{1}{8} e^{-1} - \frac{1}{24} e^{-3}$$

$$30. \int_{-\infty}^{\infty} \frac{x e^{ix}}{(x^2+1)(x^2+4)} dx = 2\pi i [\text{Res}(f(z), i) + \text{Res}(f(z), 2i)] = 2\pi i \left[\frac{1}{6} e^{-1} - \frac{1}{6} e^{-2} \right] = \frac{\pi}{3} (e^{-1} - e^{-2})i;$$

$$\int_{-\infty}^{\infty} \frac{x \sin x}{(x^2+1)(x^2+4)} dx = \text{Im} \left(\int_{-\infty}^{\infty} \frac{x e^{ix}}{(x^2+1)(x^2+4)} dx \right) = \frac{\pi}{3} (e^{-1} - e^{-2}).$$

Therefore,

$$\int_0^{\infty} \frac{x \sin x}{(x^2+1)(x^2+4)} dx = \frac{1}{2} \left[\frac{\pi}{3} (e^{-1} - e^{-2}) \right] = \frac{\pi}{6} (e^{-1} - e^{-2}).$$

31. Consider the contour integral $\oint_C \frac{e^{iz}}{z} dz$. The function $f(z) = \frac{1}{z}$ has a simple pole at $z = 0$. If we use the contour C shown in Figure 19.14, it follows from the Cauchy-Goursat Theorem that

$$\oint_C = \int_{C_R} + \int_{-R}^{-r} + \int_{-C_r} + \int_r^R = 0.$$

Taking limits as $R \rightarrow \infty$ and as $r \rightarrow 0$ and using Theorem 19.17 we then find

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx - \pi i \text{Res}(f(z)e^{iz}, 0) = 0 \quad \text{or} \quad \text{P.V.} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx = \pi i.$$

Equating the imaginary parts of $\int_{-\infty}^{\infty} \frac{\cos x + i \sin x}{x} dx = 0 + \pi i$ gives

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} dx = \pi.$$

32. Consider the contour integral $\oint_C \frac{e^{iz}}{z(z^2+1)} dz$. The function $f(z) = \frac{1}{z(z^2+1)}$ has simple poles at $z = 0$ and at $z = i$. If we use the contour C shown in Figure 19.14, it follows from Theorem 19.14 that

$$\oint_C = \int_{C_R} + \int_{-R}^{-r} + \int_{-C_r} + \int_r^R = 2\pi i \text{Res}(f(z), i).$$

Taking limits as $R \rightarrow \infty$ and as $r \rightarrow 0$ and using Theorem 19.17 we then find

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2+1)} dx - \pi i \text{Res}(f(z)e^{iz}, 0) = 2\pi i \text{Res}(f(z)e^{iz}, i)$$

or

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2+1)} dx = \pi i + 2\pi i \left(-\frac{e^{-1}}{2} \right).$$

Equating the imaginary parts of $\int_{-\infty}^{\infty} \frac{\cos x + i \sin x}{x(x^2+1)} dx = 0 + \pi(1 - e^{-1})i$ gives

$$\int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+1)} dx = \pi(1 - e^{-1}).$$

$$33. \int_0^{\pi} \frac{d\theta}{(a + \cos \theta)^2} = \frac{1}{2} \int_0^{2\pi} \frac{d\theta}{(a + \cos \theta)^2} = \frac{2}{i} \oint_C \frac{z}{(z^2 + 2az + 1)^2} dz \quad (C \text{ is } |z| = 1) = \frac{2}{i} \oint_C \frac{z}{(z - r_1)^2(z - r_2)^2} dz$$

where $r_1 = -a + \sqrt{a^2 - 1}$, $r_2 = -a - \sqrt{a^2 - 1}$. Now

$$\oint_C \frac{z}{(z - r_1)^2(z - r_2)^2} dz = 2\pi i \text{Res}(f(z), r_1) = 2\pi i \frac{a}{4(\sqrt{a^2 - 1})^3} = \frac{a\pi}{2(\sqrt{a^2 - 1})^3} i.$$

19.6 Evaluation of Real Integrals

Thus,

$$\int_0^\pi \frac{d\theta}{(a + \cos \theta)^2} = \frac{2}{i} \cdot \frac{a\pi}{2(\sqrt{a^2 - 1})^3} i = \frac{a\pi}{(\sqrt{a^2 - 1})^3}.$$

When $a = 2$ we obtain

$$\int_0^\pi \frac{d\theta}{(2 + \cos \theta)^2} = \frac{2\pi}{(\sqrt{3})^3} \quad \text{and so} \quad \int_0^{2\pi} \frac{d\theta}{(2 + \cos \theta)^2} = \frac{4\pi}{3\sqrt{3}}.$$

34. $\int_0^{2\pi} \frac{\sin^2 \theta}{a + b \cos \theta} d\theta = \frac{i}{2b} \oint_C \frac{z^2 - 1}{z^2(z - r_1)(z - r_2)} dz$ (C is $|z| = 1$) where $r_1 = (-a + \sqrt{a^2 - b^2})/b$, $r_2 = (-a - \sqrt{a^2 - b^2})/b$. Now

$$\oint_C \frac{z^2 - 1}{z^2(z - r_1)(z - r_2)} dz = 2\pi i [\text{Res}(f(z), 0) + \text{Res}(f(z), r_1)] = 2\pi i \left[-\frac{2a}{b} + \frac{2\sqrt{a^2 - b^2}}{b} \right].$$

Thus,

$$\int_0^{2\pi} \frac{\sin^2 \theta}{a + b \cos \theta} d\theta = \frac{2\pi}{b^2} (a - \sqrt{a^2 - b^2}), \quad a > b > 0.$$

When $a = 5$, $b = 4$ we obtain

$$\int_0^{2\pi} \frac{\sin^2 \theta}{5 + 4 \cos \theta} d\theta = \frac{2\pi}{16} (5 - \sqrt{9}) = \frac{\pi}{4}.$$

35. Consider the contour integral $\oint_C \frac{e^{az}}{1 + e^z} dz$. The function $f(z) = \frac{e^{az}}{1 + e^z}$ has simple poles at $z = \pi i, 3\pi i, 5\pi i, \dots$ in the upper plane. Using the contour in Figure 19.15 we have from Theorem 19.14

$$\oint_C = \int_{-r}^r + \int_{C_2} + \int_{C_3} + \int_{C_4} = 2\pi i \text{Res}(f(z), \pi i) = -2\pi i e^{a\pi i}.$$

On C_2 , $z = r + iy$, $0 \leq y \leq \pi$, $dz = i dy$,

$$\left| \int_{C_2} \frac{e^{az}}{1 + e^z} dz \right| \leq \frac{e^{ar}}{e^r - 1} (2\pi).$$

Because $0 < a < 1$, this last expression goes to 0 as $r \rightarrow \infty$. On C_3 , $z = x + 2\pi i$, $-r \leq x \leq r$, $dz = dx$,

$$\int_{C_3} \frac{e^{az}}{1 + e^z} dz = \int_r^{-r} \frac{e^{a(x+2\pi i)}}{1 + e^{x+2\pi i}} dx = -e^{2a\pi i} \int_{-r}^r \frac{e^{ax}}{1 + e^x} dx.$$

On C_4 , $z = -r + iy$, $0 \leq y \leq 2\pi$, $dz = i dy$,

$$\left| \int_{C_4} \frac{e^{az}}{1 + e^z} dz \right| \leq \frac{e^{-ar}}{1 - e^{-r}} (2\pi).$$

Because $0 < a$, this last expression goes to 0 as $r \rightarrow \infty$. Hence

$$\int_{-r}^r \frac{e^{ax}}{1 + e^x} dx - e^{2a\pi i} \int_{-r}^r \frac{e^{ax}}{1 + e^x} dx = -2\pi i e^{a\pi i}$$

gives, as $r \rightarrow \infty$,

$$(1 - e^{2a\pi i}) \int_{-\infty}^{\infty} \frac{e^{ax}}{1 + e^x} dx = -2\pi i e^{a\pi i}.$$

That is

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1 + e^x} dx = -\frac{2\pi i e^{a\pi i}}{1 - e^{2a\pi i}} = \frac{\pi}{\frac{e^{a\pi i} - e^{-a\pi i}}{2i}} = \frac{\pi}{\sin ax}.$$

36. Using the Fourier sine transform with respect to y the partial differential equation becomes $\frac{d^2 U}{dx^2} - \alpha^2 U = 0$ and so

$$U(x, \alpha) = c_1 \cosh \alpha x + c_2 \sinh \alpha x.$$

The boundary condition $u(0, y)$ becomes $U(0, \alpha) = 0$ and so $c_1 = 0$. Thus $U(x, \alpha) = c_2 \sinh \alpha x$. Now to evaluate

$$U(\pi, \alpha) = \int_0^\infty \frac{2y}{y^4 + 4} \sin \alpha y \, dy = \int_{-\infty}^\infty \frac{y}{y^4 + 4} \sin \alpha y \, dy$$

we use the contour integral $\int_C \frac{ze^{i\alpha z}}{z^4 + 4} dz$ and

$$\int_{-\infty}^\infty \frac{xe^{i\alpha x}}{x^4 + 4} dx = 2\pi i [\text{Res}(f(z), 1+i) + \text{Res}(f(z), -1+i)] = 2\pi i \left[-\frac{1}{8}ie^{(-1+i)\alpha} + \frac{1}{8}ie^{(-1-i)\alpha} \right] = \frac{\pi}{2}(e^{-\alpha} \sin \alpha)i$$

$$\int_{-\infty}^\infty \frac{x \sin \alpha x}{x^4 + 4} dx = \text{Im} \left(\int_{-\infty}^\infty \frac{xe^{i\alpha x}}{x^4 + 4} dx \right) = \frac{\pi}{2}e^{-\alpha} \sin \alpha.$$

Finally, $U(\pi, \alpha) = \frac{\pi}{2}e^{-\alpha} \sin \alpha = c_2 \sinh \alpha \pi$ gives $c_2 = \frac{\pi}{2} \frac{e^{-\alpha} \sin \alpha}{\sinh \alpha \pi}$. Hence $U(x, \alpha) = \frac{\pi}{2} \frac{e^{-\alpha} \sin \alpha}{\sinh \alpha \pi} \sinh \alpha x$ and

$$u(x, y) = \int_0^\infty \frac{e^{-\alpha} \sin \alpha}{\sinh \alpha \pi} \sinh \alpha x \sin \alpha y \, d\alpha.$$

CHAPTER 19 REVIEW EXERCISES

1. True
 2. False
 3. False
 4. True
 5. True
 6. True
 7. True
 8. five
 9. $1/\pi$
 10. three; $-1/6$
 11. $|z - i| = \sqrt{5}$
 12. False
 13.
$$\frac{e^{z(1+i)} + e^{z(1-i)}}{2} = \frac{1}{2} \left(1 + z(1+i) + \frac{z^2}{2!}(1+i)^2 + \cdots \right) + \frac{1}{2} \left(1 + z(1-i) + \frac{z^2}{2!}(1-i)^2 + \cdots \right)$$

$$= 1 + z \left[\frac{(1+i) + (1-i)}{2} \right] + \frac{z^2}{2!} \left[\frac{(1+i)^2 + (1-i)^2}{2} \right] + \cdots = 1 + \sum_{k=1}^\infty \frac{(\sqrt{2})^k \cos \frac{k\pi}{4}}{k!} z^k$$
- Here we have used $(1+i)^n = (\sqrt{2})^n e^{n\pi i/4}$ and $(1-i)^n = (\sqrt{2})^n e^{-n\pi i/4}$ so that
- $$\frac{(1+i)^n + (1-i)^n}{2} = (\sqrt{2})^n \left[\frac{e^{n\pi i/4} + e^{-n\pi i/4}}{2} \right] = (\sqrt{2})^n \cos \frac{n\pi}{4}.$$
14. $\sin \frac{\pi}{z} = 0$ implies $z = \frac{1}{n}$, $n = \pm 1, \pm 2, \dots$. All singularities are isolated except the singularity $z = 0$.
 15. $f(z) = \frac{1}{z^4} \left[1 - \left(1 + \frac{iz}{1!} + \frac{i^2 z^2}{2!} + \frac{i^3 z^3}{3!} + \frac{i^4 z^4}{4!} + \cdots \right) \right] = -\frac{i}{z^3} + \frac{1}{2!z^2} + \frac{i}{3!z} - \frac{1}{4!} - \frac{iz}{5!} + \cdots$

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$$16. e^{z/(z-2)} = e \cdot e^{2/(z-2)} = e \left(1 + \frac{2}{z-2} + \frac{2^2}{2!(z-2)^2} + \frac{2^3}{3!(z-2)^3} + \cdots \right) = e \sum_{k=0}^{\infty} \frac{2^k}{k!} (z-2)^{-k}$$

$$17. (z-i)^2 \sin \frac{1}{z-i} = (z-i)^2 \left[\frac{1}{z-i} - \frac{1}{3!(z-i)^3} + \frac{1}{5!(z-i)^5} - \cdots \right] = \cdots + \frac{1}{5!(z-i)^3} - \frac{1}{3!(z-i)} + (z-i)$$

$$18. \frac{1 - \cos z^2}{z^5} = \frac{1}{z^5} \left[1 - \left(1 - \frac{z^4}{2!} + \frac{z^8}{4!} - \frac{z^{12}}{6!} + \frac{z^{16}}{8!} - \cdots \right) \right] = \frac{1}{2!z} - \frac{z^3}{4!} + \frac{z^7}{6!} - \frac{z^{11}}{8!} + \cdots$$

$$19. (a) f(z) = \frac{1}{z-3} - \frac{1}{z-1} = \frac{1}{1-z} - \frac{1}{3} \cdot \frac{1}{1-\frac{z}{3}} = (1+z+z^2+z^4+\cdots) - \frac{1}{3} \left(1 + \frac{z}{3} + \frac{z^2}{3^2} + \frac{z^3}{3^3} + \cdots \right)$$

$$= \frac{2}{3} + \frac{8}{9}z + \frac{26}{27}z^2$$

$$(b) f(z) = -\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} - \frac{1}{3} \cdot \frac{1}{1-\frac{z}{3}} = -\frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots \right) - \frac{1}{3} \left(1 + \frac{z}{3} + \frac{z^2}{3^2} + \frac{z^3}{3^3} + \cdots \right)$$

$$= \cdots - \frac{1}{z^3} - \frac{1}{z^2} - \frac{1}{z} - \frac{1}{3} - \frac{z}{3^2} - \frac{z^2}{3^3} - \cdots$$

$$(c) f(z) = -\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} + \frac{1}{z} \cdot \frac{1}{1-\frac{z}{3}} = -\frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots \right) + \frac{1}{z} \left(1 + \frac{z}{3} + \frac{z^2}{3^2} + \frac{z^3}{3^3} + \cdots \right)$$

$$= \frac{2}{z^2} + \frac{8}{z^3} + \frac{26}{z^4} + \cdots$$

$$(d) f(z) = -\frac{1}{z-1} - \frac{1}{2} \cdot \frac{1}{1-\frac{z-1}{2}} = -\frac{1}{z-1} - \frac{1}{2} \left(1 + \frac{z-1}{2} + \frac{(z-1)^2}{2^2} + \frac{(z-1)^3}{3!} + \cdots \right)$$

$$= -\frac{1}{z-1} - \frac{1}{2} - \frac{z-1}{2^2} - \frac{(z-1)^2}{2^3} - \cdots$$

$$20. (a) f(z) = \frac{1}{25} \left(1 - \frac{z}{5} \right)^{-2} = \frac{1}{25} \left[1 + (-2) \left(-\frac{z}{5} \right) + \frac{(-2)(-3)}{2!} \left(-\frac{z}{5} \right)^2 + \frac{(-2)(-3)(-4)}{3!} \left(-\frac{z}{5} \right)^3 + \cdots \right]$$

$$= \frac{1}{25} + 2 \frac{z}{5^3} + 3 \frac{z^2}{5^4} + 4 \frac{z^3}{5^5} + \cdots$$

$$(b) (z-5)^{-2} = \frac{1}{z^2} \left(1 - \frac{5}{z} \right)^{-2} = \frac{1}{z^2} \left[1 + (-2) \left(-\frac{5}{z} \right) + \frac{(-2)(-3)}{2!} \left(-\frac{5}{z} \right)^2 + \frac{(-2)(-3)(-4)}{3!} \left(-\frac{5}{z} \right)^3 + \cdots \right]$$

$$= \frac{1}{z^2} + 2 \frac{5}{z^3} + 3 \frac{5^2}{z^4} + 4 \frac{5^3}{z^5} + \cdots$$

$$(c) \frac{1}{(z-5)^2} \text{ is the Laurent series.}$$

$$21. \oint_C \frac{2z+5}{z(z+2)(z-1)^4} dz = 2\pi i [\text{Res}(f(z), 0) + \text{Res}(f(z), -2)] = \frac{404}{81} \pi i$$

$$22. \oint_C \frac{z^2}{(z-1)^3(z^2+4)} dz = 2\pi i \text{Res}(f(z), 1) = \frac{8\pi}{125} i$$

$$23. \oint_C \frac{1}{2 \sin z - 1} dz = 2\pi i \text{Res} \left(f(z), \frac{\pi}{6} \right) = \frac{2\pi}{\sqrt{3}} i$$

$$24. \oint_C \frac{z+1}{\sinh z} dz = 2\pi i [\text{Res}(f(z), 0) + \text{Res}(f(z), \pi i)] = 2\pi i [1 + (-\pi i - 1)] = 2\pi^2$$

$$\begin{aligned}
 25. \quad \oint_C \frac{e^{2z}}{z^4 + 2z^3 + 2z^2} dz &= 2\pi i [\operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), -1 + i) + \operatorname{Res}(f(z), -1 - i)] \\
 &= 2\pi i \left[\frac{1}{2} + \frac{e^{-2}}{4}(\cos 2 + i \sin 2) + \frac{e^{-2}}{4}(\cos 2 - i \sin 2) \right] = \pi(1 + e^{-2} \cos 2)i
 \end{aligned}$$

$$26. \quad \oint_C \frac{1}{z^4 - 2z^2 + 4} dz = 2\pi i \left[\operatorname{Res} \left(f(z), \frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2}i \right) + \operatorname{Res} \left(f(z), -\frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2}i \right) \right] = \frac{\pi}{2\sqrt{2}}$$

$$27. \quad \oint_C \frac{1}{z(e^z - 1)} dz = 2\pi i \operatorname{Res}(f(z), 0) = -\pi i. \quad \text{Note: } z = 0 \text{ is a pole of order two, and so}$$

$$\operatorname{Res}(f(z), 0) = \lim_{z \rightarrow 0} \frac{d}{dz} z^2 \cdot \frac{1}{z^2 \left(1 + \frac{z}{2!} + \frac{z^2}{3!} + \cdots \right)} = \lim_{z \rightarrow 0} - \frac{\left(\frac{1}{2!} + \frac{2z}{3!} + \cdots \right)}{\left(1 + \frac{z}{2!} + \frac{z^2}{3!} + \cdots \right)^2} = -\frac{1}{2}.$$

$$28. \quad \oint_C \frac{z}{(z-1)(z+1)^{10}} dz = 2\pi i [\operatorname{Res}(f(z), 1) + \operatorname{Res}(f(z), -1)] = 2\pi i \left[\frac{1}{2^{10}} + \left(-\frac{1}{2^{10}} \right) \right] = 0$$

29. Using two integrals,

$$\begin{aligned}
 \oint_C z e^{3/z} dz + \oint_C \frac{\sin z}{z^2(z-\pi)^3} dz &= 2\pi i \operatorname{Res}(f(z), 0) + 2\pi i [\operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), \pi)] \\
 &= 2\pi i \cdot \frac{9}{2} + 2\pi i \left[-\frac{1}{\pi^3} + \frac{2}{\pi^3} \right] = \left(9\pi + \frac{2}{\pi^2} \right) i.
 \end{aligned}$$

Note: In the first integral $z = 0$ is an essential singularity and the residue is obtained from the Laurent series

$$z e^{3/z} = \cdots + \frac{3^3}{3!z^2} + \frac{3^2}{2!z} + 3 + z.$$

$$30. \quad \oint_C \csc \pi z dz = 2\pi i [\operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), 1) + \operatorname{Res}(f(z), 2)] = 2\pi i \left[\frac{1}{\pi} + \left(-\frac{1}{\pi} \right) + \frac{1}{\pi} \right] = 2i$$

$$31. \quad \int_{-\infty}^{\infty} \frac{x^2}{(x^2 + 2x + 2)(x^2 + 1)^2} dx = 2\pi i [\operatorname{Res}(f(z), -1 + i) + \operatorname{Res}(f(z), i)] = 2\pi i \left[\frac{3}{25} - \frac{4}{25}i - \frac{3}{25} + \frac{9}{100}i \right] = \frac{7\pi}{50}$$

$$32. \quad \int_{-\infty}^{\infty} \frac{x + ai}{x^2 + a^2} e^{ix} dx = \int_{-\infty}^{\infty} \frac{x \cos x - a \sin x}{x^2 + a^2} dx + i \int_{-\infty}^{\infty} \frac{a \cos x + x \sin x}{x^2 + a^2} dx = 0 + 2\pi i \operatorname{Res}(f(z), ai) = 2\pi i e^{-a}$$

Thus,

$$\int_{-\infty}^{\infty} \frac{a \cos x + x \sin x}{x^2 + a^2} dx = 2\pi e^{-a}.$$

$$\begin{aligned}
 33. \quad \int_0^{2\pi} \frac{\cos^2 \theta}{2 + \sin \theta} d\theta &= \frac{1}{2} \oint_C \frac{z^4 + 2z^2 + 1}{z^2(z^2 + 4iz - 1)} dz \quad (C \text{ is } |z| = 1) = \pi i [\operatorname{Res}(f(z), 0) + \operatorname{Res}(f(z), (-2 + \sqrt{3})i)] \\
 &= \pi i [-4i + 2\sqrt{3}i] = (4 - 2\sqrt{3})\pi
 \end{aligned}$$

[Note: The answer in the text is correct but not simplified.]

$$\begin{aligned}
 34. \quad \int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} d\theta &= -\frac{1}{2i} \oint_C \frac{z^6 + 1}{z^3(z-2)(2z-1)} dz \quad (C \text{ is } |z| = 1) = -\pi \left[\operatorname{Res}(f(z), 0) + \operatorname{Res} \left(f(z), \frac{1}{2} \right) \right] \\
 &= -\pi \left[\frac{21}{8} + \left(-\frac{65}{24} \right) \right] = \frac{\pi}{12}
 \end{aligned}$$

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35. The integrand of $\oint_C \frac{1 - e^{iz}}{z^2} dz$ has a simple pole at $z = 0$. Using a contour as in Figure 19.14 of Section 19.6 we have

$$\oint_C = \int_{C_R} + \int_{-R}^{-r} + \int_{-C_R} + \int_r^R = 0.$$

By taking limits as $R \rightarrow 0$ and as $r \rightarrow 0$, and using Theorem 19.17 we find

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{1 - e^{ix}}{x^2} dx - \pi i \text{Res}(f(z), 0) = 0.$$

Thus,

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{1 - \cos x - i \sin x}{x^2} dx = \pi.$$

Equating real parts gives

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{1 - \cos x}{x^2} dx = \pi.$$

Finally,

$$\int_0^{\infty} \frac{1 - \cos x}{x^2} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{1 - \cos x}{x^2} dx = \frac{\pi}{2}.$$

36. We have

$$C e^{-a^2 z^2} e^{ibz} dz = \int_{-r}^r + \int_{C_1} + \int_{C_2} + \int_{C_3} = 0$$

by the Cauchy-Goursat Theorem. Therefore,

$$\int_{-r}^r = - \int_{C_1} - \int_{C_2} - \int_{C_3}.$$

Let C_1 and C_3 denote the vertical sides of the rectangle. By the ML-inequality, $\int_{C_1} \rightarrow 0$ and $\int_{C_3} \rightarrow 0$ as $r \rightarrow \infty$. On C_2 , $z = x + \frac{b}{2a^2}i$, $-r \leq x \leq r$, $dz = dx$,

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-ax^2} e^{ibx} dx &= - \int_{-\infty}^{\infty} e^{-a^2(x + \frac{b}{2a^2}i)^2} e^{ib(x + \frac{b}{2a^2}i)} dx = \int_{-\infty}^{\infty} e^{-a^2 x^2} e^{-b^2/4a^2} dx \\ &= \int_{-\infty}^{\infty} e^{-a^2 x^2} (\cos bx + i \sin bx) dx = e^{-b^2/4a^2} \int_{-\infty}^{\infty} e^{-a^2 x^2} dx. \end{aligned}$$

Using the given value of $\int_{-\infty}^{\infty} e^{-a^2 x^2} dx$ and equating real and imaginary parts gives

$$\int_{-\infty}^{\infty} e^{-a^2 x^2} \cos bx dx = \frac{\sqrt{\pi}}{a} e^{-b^2/4a^2} \quad \text{and so} \quad \int_0^{\infty} e^{-a^2 x^2} \cos bx dx = \frac{\sqrt{\pi}}{2a} e^{-b^2/4a^2}.$$

37.
$$\begin{aligned} a_k &= \frac{1}{2\pi i} \oint_C \frac{e^{(u/2)(z-z^{-1})}}{z^{k+1}} dz = \frac{1}{2\pi i} \int_0^{2\pi} \frac{e^{(u/2)(e^{it}-e^{-it})}}{(e^{it})^{k+1}} i e^{it} dt = \frac{1}{2\pi} \int_0^{2\pi} e^{(u/2)(2i \sin t)} e^{-kit} dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} e^{-i(kt-u \sin t)} dt = \frac{1}{2\pi} \int_0^{2\pi} [\cos(kt - u \sin t) - i \sin(kt - u \sin t)] dt = \frac{1}{2\pi} \int_0^{2\pi} \cos(kt - u \sin t) dt \end{aligned}$$

since $\int_0^{2\pi} \sin(kt - u \sin t) dt = 0$. (To obtain this last result, expand the integrand and let $t = 2\pi - x$.)